Unmanned combat aerial vehicles (UCAVs) have further extended drone functionality to include precision-guided strikes, minimizing collateral damage and reducing the exposure of human pilots to hostile zones. Drones also play an essential role in logistics, transporting supplies, conducting damage assessments, and supporting humanitarian operations in conflict zones. Additionally, swarm drone systems and AI-enabled autonomous coordination are emerging as critical assets in electronic warfare, decoy operations, and tactical defense scenarios.

Figure 1.1 illustrates the diverse and strategic roles of drones in mil-itary applications, emphasizing their transformation into critical assets for modern defense operations. The figure highlights how unmanned aerial vehi-cles (UAVs) are deployed in surveillance, reconnaissance, combat, logistics, and intelligence missions, enhancing battlefield awareness and operational efficiency. Equipped with high-resolution cameras, thermal sensors, and communication modules, drones conduct real-time surveillance and gather crucial intelligence, supporting rapid tactical decisions. Armed variants, known as unmanned combat aerial vehicles (UCAVs), perform precision strikes on high-value targets while minimizing pilot risk. Additionally, drones are utilized for logistics support, delivering essential supplies to inaccessible or dangerous zones, and for search and rescue (SAR) missions, aiding in locating personnel and assessing damage in post-conflict environments.